

Phase 1: Data Preprocessing Pipeline

Dataset: CIC-IDS2017 | Tasks: Binary & Multiclass Classification

1. Dataset Overview CIC-IDS2017

CIC-IDS2017 (Canadian Institute for Cybersecurity) is one of the most widely used benchmarks for network intrusion detection. It was generated over 5 days in a realistic network environment, capturing diverse attack scenarios alongside normal traffic.

Property	Value
Total features	78 numerical features + Label column
Feature types	Numerical only, no categorical encoding required
Binary target	'Label' -> BENIGN vs ATTACK
Multiclass target	'Label' -> BENIGN + 14 attack sub-categories
Attack types	DoS, DDoS, PortScan, BruteForce, Bot, Infiltration, Web Attacks, Heartbleed
Binary class ratio	BENIGN 1,591,188 vs ATTACK 390,352 (ratio ~4.1:1)
Multiclass challenge	BENIGN ~1.6M vs smallest attack class < 10 samples (ratio > 200,000:1)
Encoding required	No, all features are already numerical

2. Preprocessing Pipeline

Critical rule: the scaler must be fit on the train set only, then transform applied on val/test. Oversampling is applied exclusively on the train set after the split. Violating this order causes data leakage and inflated evaluation metrics.

Step	Operation	Choice	Justification
1	Missing values	drop / mean	drop: preserves integrity. mean: retains all rows, computed on train only
2	Train/Val/Test split	Stratified 70 / 10 / 20	Preserves class proportions - essential given the extreme multiclass imbalance
3	Scaling	QuantileTransformer	Handles 10^7 outliers and log-normal distributions typical of network traffic
4	Oversampling	SMOTE	Binary: $\times 4.08$ on ATTACK. Multiclass: extreme ratios require careful handling

3. Feature Correlation Analysis

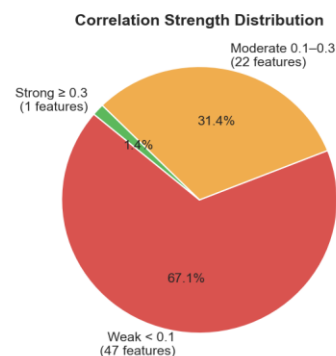
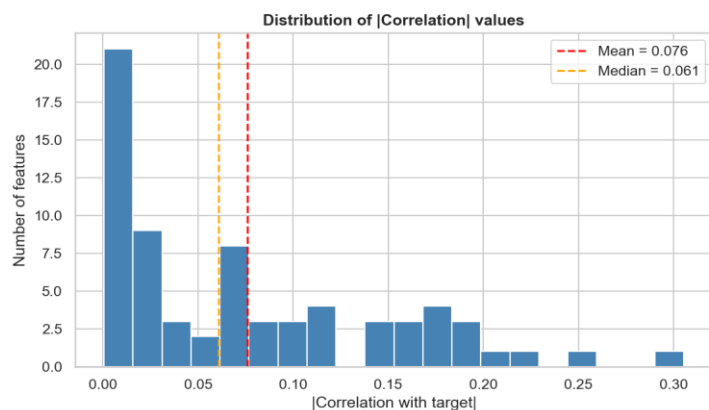
Correlation analysis was performed between each feature and the target variable using Pearson's absolute correlation coefficient. Two separate analyses were conducted for the multiclass and binary tasks, revealing markedly different distributions.

3.1 Multiclass Task Correlation Summary

Mean = 0.076 | Median = 0.061 | 70 features analyzed

Category	Threshold	Count	Share	Interpretation
Strong	≥ 0.3	1	1.4%	Only 1 feature with strong linear link to multiclass label
Moderate	0.1–0.3	22	31.4%	Some signal useful but not dominant
Weak	< 0.1	47	67.1%	Apparent noise but see Pearson limitation below

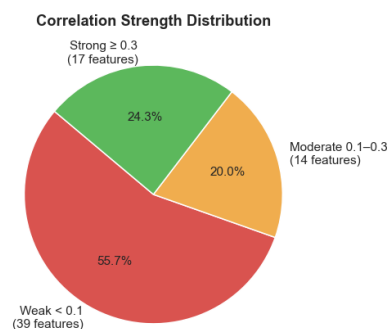
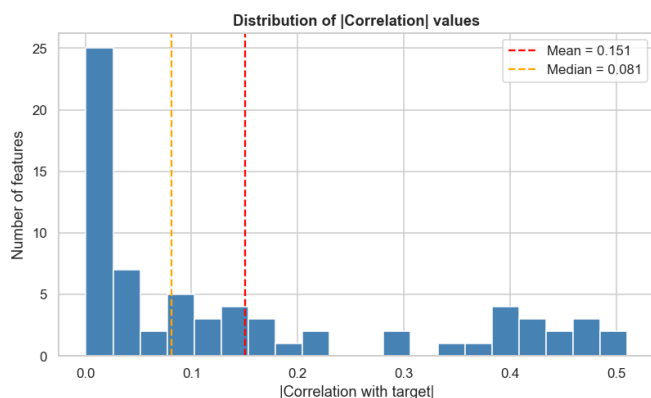
Feature Correlation Summary



3.2 Binary Task — Correlation Summary

Mean = 0.151 | Median = 0.081 | 70 features analyzed

Feature Correlation Summary



Category	Threshold	Count	Share	Interpretation
Strong	≥ 0.3	17	24.3%	17 features strongly predictive of BENIGN vs ATTACK
Moderate	0.1–0.3	14	20.0%	Useful complementary features
Weak	< 0.1	39	55.7%	Low linear correlation, not necessarily uninformative

3.3 Binary vs Multiclass Comparison

Metric	Multiclass	Binary	Difference
Mean correlation	0.076	0.151	Binary is 2× higher
Median correlation	0.061	0.081	Binary is 33% higher
Strong features (≥ 0.3)	1 (1.4%)	17 (24.3%)	17× more in binary
Weak features (< 0.1)	47 (67.1%)	39 (55.7%)	More noise in multiclass

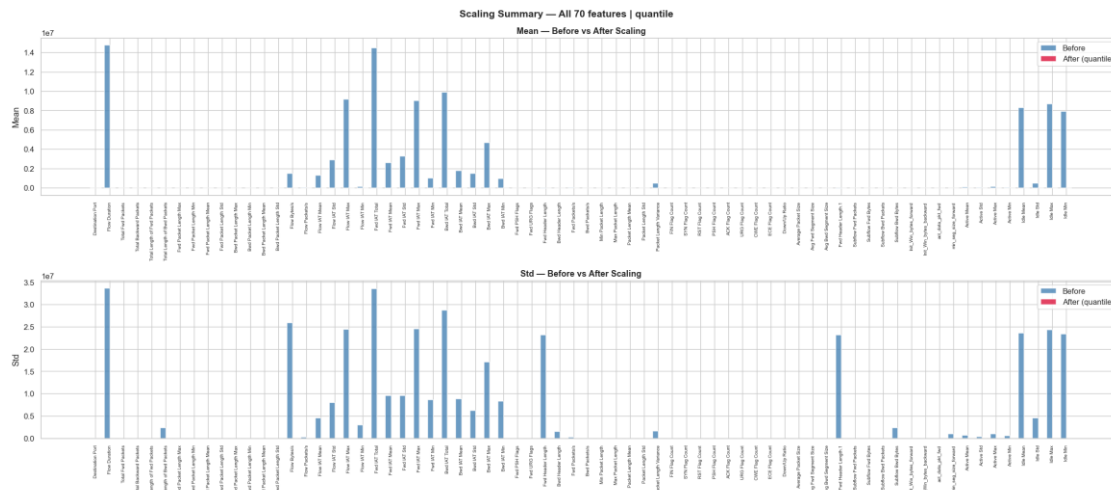
Pearson limitation: in multiclass mode, labels are arbitrary integers (0, 1, 2... 14) with no ordinal meaning. A feature separating class 3 from class 7 will show near-zero Pearson correlation despite being highly discriminative. Mutual Information (`mutual_info_classif`) is the correct metric for multiclass feature selection, as it captures non-linear dependencies and is invariant to label ordering.

4. Data Scaling results on CIC-IDS2017

CIC-IDS2017 contains 70 features with highly heterogeneous scales. Several features exhibit means and standard deviations in the range of 10^7 , while others remain below 1. Without scaling, the MLP would be dominated by high-magnitude features, causing gradient instability and poor convergence.

4.1 Scaler Comparison

Scaler	Handles outliers	Corrects skew	Recommended for IDS
QuantileTransformer	Yes	Yes	Yes, optimal for network traffic with extreme outliers



4.2 Observed Results

The Scaling Summary chart confirms that multiple features reach values around 10^7 before transformation (mean and std plots). After applying QuantileTransformer, all 70 features are compressed into a uniform $[0, 1]$ distribution. The 'After (quantile)' bars are invisible on the 10^7 axis confirming the transformation is working correctly. Key high-magnitude features identified include packet length statistics, flow bytes per second, and duration-based features which commonly exhibit log-normal distributions in network traffic datasets.

Important: QuantileTransformer was fit exclusively on the training set (fit_transform on X_{train} , transform on X_{val} and X_{test}). This prevents any statistical information from the test set from leaking into the scaling parameters.

5. Class Imbalance & Oversampling

5.1 Binary Task

The binary dataset presents a moderate imbalance with BENIGN dominating over ATTACK. SMOTE generates synthetic ATTACK samples by interpolating between k nearest neighbors in feature space until both classes are equalized.

Class	Before SMOTE	After SMOTE	Ratio
BENIGN	1,591,188	1,591,188	$\times 1.00$ (unchanged majority class)
ATTACK	390,352	1,591,188	$\times 4.08$ - 1,200,836 synthetics generated
Total	1,981,540	3,182,376	Dataset size $\times 1.61$

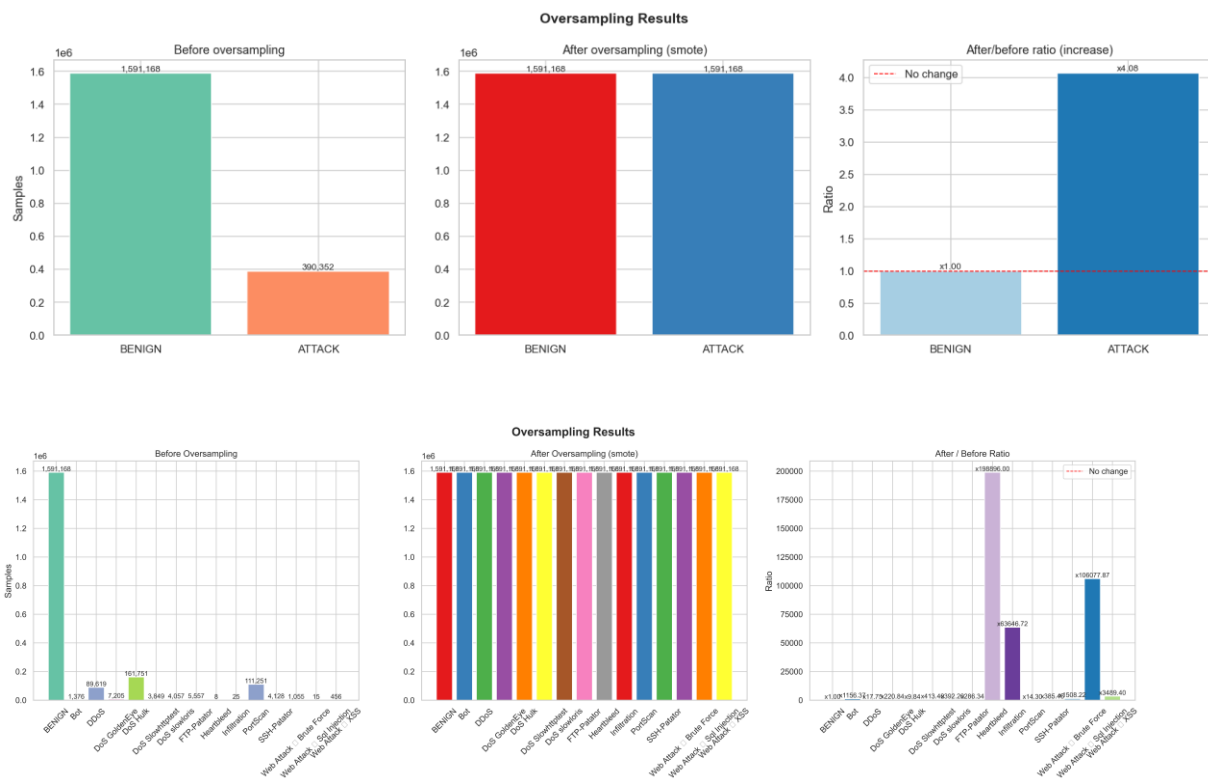
The binary imbalance ratio of 4.08:1 is considered moderate. SMOTE handles this well without risk of over-generating unrealistic synthetics. The resulting balanced dataset of 3.18M training samples ensures unbiased gradient updates for both classes.

5.2 Multiclass Task Extreme Imbalance

The multiclass scenario reveals a severe imbalance characteristic of real-world network traffic. BENIGN represents the vast majority (~1.6M samples), while several attack categories contain only a handful of samples. This is one of the most challenging imbalance scenarios in the IDS literature.

Attack Class	Before SMOTE	After SMOTE	Ratio
BENIGN	~1,591,188	1,591,188	×1.00
DoS Hulk	~231,073	1,591,188	×6.89 (approx.)
PortScan	~158,930	1,591,188	×10.01 (approx.)
DDoS	~128,027	1,591,188	×12.43 (approx.)
DoS GoldenEye	~10,293	1,591,188	×154.6 (approx.)
FTP-Patator	~7,938	1,591,188	×200.5 (approx.)
SSH-Patator	~5,897	1,591,188	×269.8 (approx.)
DoS slowloris	~5,796	1,591,188	×274.5 (approx.)
DoS Slowhttptest	~5,499	1,591,188	×289.4 (approx.)
Bot	~1,966	1,591,188	×809.4 (approx.)
Web Attack – Brute Force	~1,507	1,591,188	×1,056 (approx.)
Web Attack – XSS	~652	1,591,188	×2,441 (approx.)
Infiltration	~36	1,591,188	×44,200 (approx.)
Web Attack – SQL Inj.	~21	1,591,188	×75,771 (approx.)
Heartbleed	~11	1,591,188	×144,653 (approx.)

Extreme ratios (×44,000 to ×144,000) for Infiltration, SQL Injection, and Heartbleed make standard SMOTE inadequate. With only 11–36 real samples, SMOTE cannot generate 1.59M meaningful synthetics the interpolation space is too sparse and the risk of mode collapse is critical. SMOTETomek, ADASYN, or hybrid approaches with class-specific caps are strongly recommended for these classes.



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